

ME 165

Basic Mechanical Engineering

Lecture 8

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Isentropic Process

- **Entropy** is the measure of the disorder of a system.
- The term "isentropic" means **constant entropy**. A process during which the entropy remains constant is called an isentropic process

$$\begin{aligned}PV^K &= \text{Constant} \\TV^{K-1} &= \text{Constant} \\TP^{\frac{1-k}{k}} &= \text{Constant} \\k &= \frac{C_p}{C_v} \\k &= \text{specific heat ratio}\end{aligned}$$

Entropy, the measure of a system's thermal energy per unit temperature that is unavailable for doing useful work. Because work is obtained from ordered molecular motion, the amount of **entropy** is also a measure of the molecular disorder, or randomness, of a system.

- **C_v** for a gas is the change in internal energy (U) of a system with respect to change in temperature at a **fixed volume** of the system i.e. $C_v = \left(\frac{\partial U}{\partial T}\right)_V$
- **C_p** for a gas is the change in the enthalpy (H) of the system with respect to change in temperature at a **fixed pressure** of the system i.e. $C_p = \left(\frac{\partial H}{\partial T}\right)_P$



Thermal Efficiency of the Ideal Diesel Cycle

$$q_{in} = P_2(v_3 - v_2) + (u_3 - u_2) \\ = h_3 - h_2 = c_p(T_3 - T_2)$$

$$q_{out} = u_4 - u_1 = c_v(T_4 - T_1)$$

$$\eta_{th,Diesel} = \frac{W_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{c_v(T_4 - T_1)}{c_p(T_3 - T_2)}$$

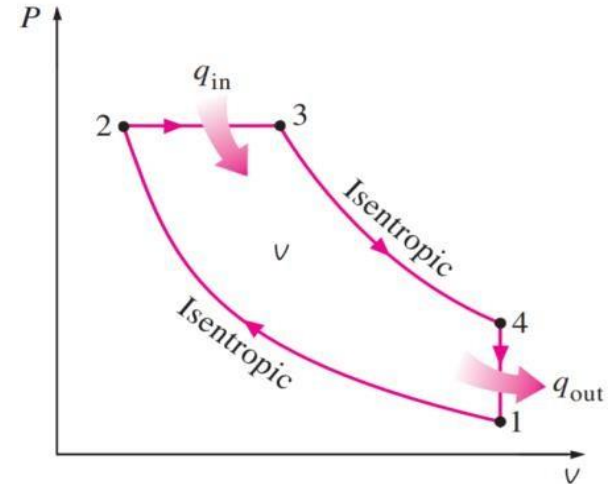
$$\therefore \eta_{th,Diesel} = 1 - \frac{T_4 - T_1}{k(T_3 - T_2)} = 1 - \frac{T_1(T_4/T_1 - 1)}{kT_2(T_3/T_2 - 1)}$$

Now, *Cutoff ratio*, $r_c = \frac{V_3}{V_2}$ and *Compression ratio*, $r = \frac{V_1}{V_2}$

Using, i) $TV^{k-1} = \text{Constant}$ for isentropic process (1-2 & 3-4)

ii) $\frac{V}{T} = \text{Constant}$ for isobaric process, iii) r_c and iv) r

$$\text{We get, } \eta_{th,Diesel} = 1 - \frac{1}{r^{k-1}} \left[\frac{r_c^k - 1}{k(r_c - 1)} \right]$$



(a) P- v diagram



Thermal Efficiency Comparison

Diesel Cycle

$$\eta_{th,Diesel} = 1 - \frac{1}{r^{k-1}} \left[\frac{r_c^k - 1}{k(r_c - 1)} \right]$$

Otto Cycle

$$\eta_{th,Otto} = 1 - \frac{1}{r^{k-1}}$$

- The efficiency of a Diesel cycle differs from the efficiency of an Otto cycle by the quantity in the brackets. This quantity is always greater than 1. Therefore, when both cycles operate on the same compression ratio, $\eta_{th,otto} > \eta_{th,Diesel}$
- Diesel engines operate at much higher compression ratios and thus are usually more efficient than the spark-ignition (gasoline) engines.
- Thermal efficiencies of large diesel engines range from about 35 to 40 percent.



Mean Effective Pressure

- It is a **theoretical parameter** used to measure the performance of an internal combustion engine (ICE).
- The mean effective pressure can be regarded as an **average pressure** in the cylinder for a complete engine cycle. By definition, mean effective pressure is the ratio between the work and engine displacement:

$$\text{MEP} = W/V_d \text{ [Pa]}$$

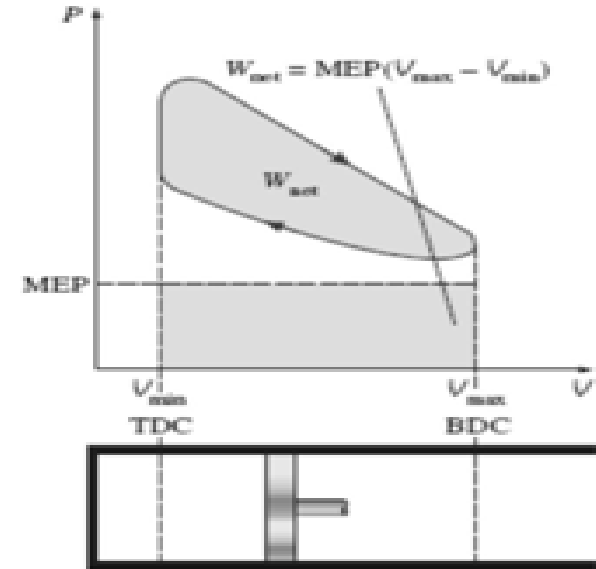
W [J] – work performed in a complete engine cycle

V_d [m³] – engine (cylinder) displacement

$$W_{\text{net}} = \text{MEP} * \text{Piston area} * \text{Stroke}$$

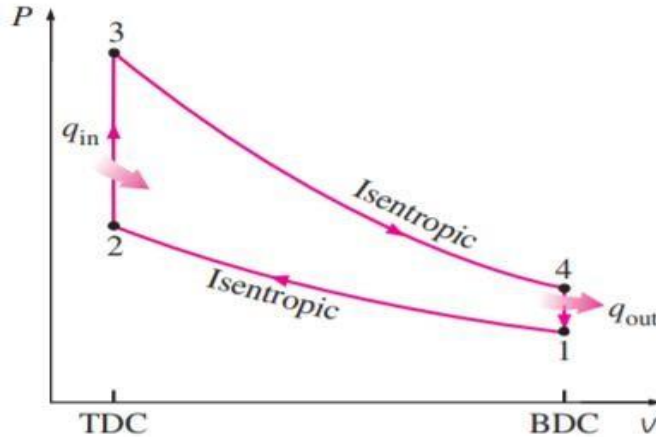
$$= \text{MEP} * \text{Displacement Volume}$$

- The MEP can be used as a parameter to **compare** the **performance** of reciprocating engines of **equal size**.
- The engine with a **larger MEP** delivers more net work per cycle and thus **better performance**.



Problem 1

In an air standard Otto cycle, the compression begins at 35 °C and 0.1 MPa and the compression ratio is 7. The maximum temperature of the cycle is 1100°C. Find (a) the temperature at various points in the cycle, (b) the heat supplied per kg of air, (c) work done per kg of air, (d) the cycle efficiency



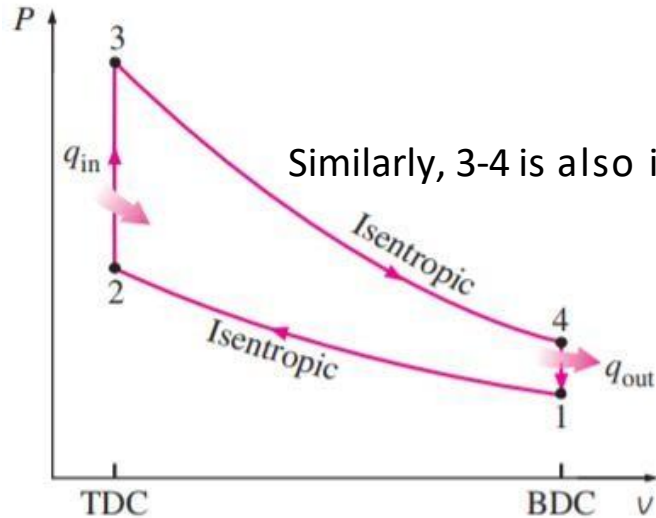
$$\begin{aligned}T_1 &= 35^\circ\text{C} = 308\text{ K} \\P_1 &= 0.1\text{ MPa} \\T_3 &= 1100^\circ\text{C} = 1373\text{ K} \\r &= v_1/v_2 = 7\end{aligned}$$

(a) At 300 K

Gas	Formula	Gas constant, R kJ/kg·K	c_p kJ/kg·K	c_v kJ/kg·K	k
Air	—	0.2870	1.005	0.718	1.400



Problem 1



Since process, 1-2 is isentropic,

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^k = 7^{1.4} = 15.24 \quad P_2 = 1524 \text{ kPa}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{k-1} = 7^{1.4-1} = 2.178 \quad T_2 = 670.8 \text{ K}$$

Similarly, 3-4 is also isentropic,

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3} \right)^{k-1} \quad T_4 = 631 \text{ K}$$

Heat rejected, $Q_{\text{out}} = C_v (T_4 - T_1) = 231.44 \text{ kJ/kg}$

Heat in, $Q_{\text{in}} = C_v (T_3 - T_2) = 504.8 \text{ kJ/kg}$

The net work output, $W_{\text{net}} = Q_{\text{in}} - Q_{\text{out}} = 272.74 \text{ kJ/kg}$

Thermal efficiency, $\eta_{\text{th,otto}} = W_{\text{net}} / Q_{\text{in}} = 0.54 = 54 \%$

Otto cycle thermal efficiency, $\eta_{\text{th,otto}} = 1 - 1/r^{k-1} = 0.54$ or 54 %



Problem 2

In a Diesel cycle, Compression begins at 0.1 MPa, 40°C and the compression ratio is 15. The heat added is 1.675 MJ/kg. Find

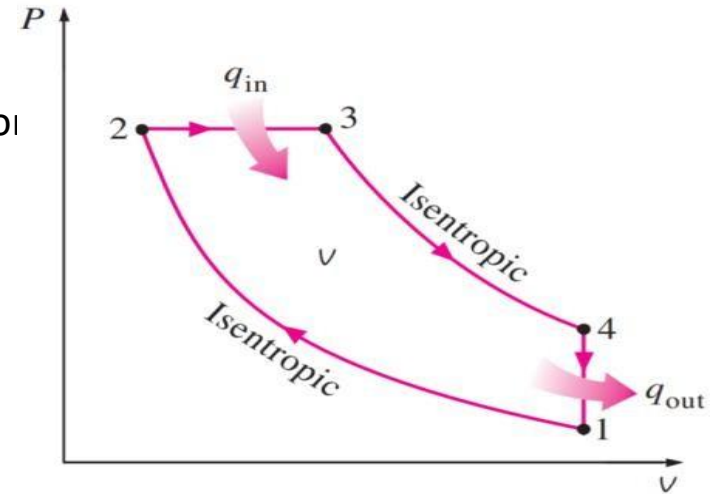
- the maximum temperature in the cycle,
- the cut-off ratio
- the temperature at the end of the isentropic expansion
- work done per kg of air
- the cycle efficiency
- the MEP of the cycle.

$$T_1 = 40^\circ\text{C} = 313\text{ K}$$

$$P_1 = 0.1\text{ MPa}$$

$$Q_{\text{in}} = 1675\text{ MJ/kg}$$

$$r = v_1/v_2 = 15$$



(a) P - v diagram

(a) At 300 K

Gas	Formula	Gas constant, R kJ/kg·K	c_p kJ/kg·K	c_v kJ/kg·K	k
Air	—	0.2870	1.005	0.718	1.400



Problem 2

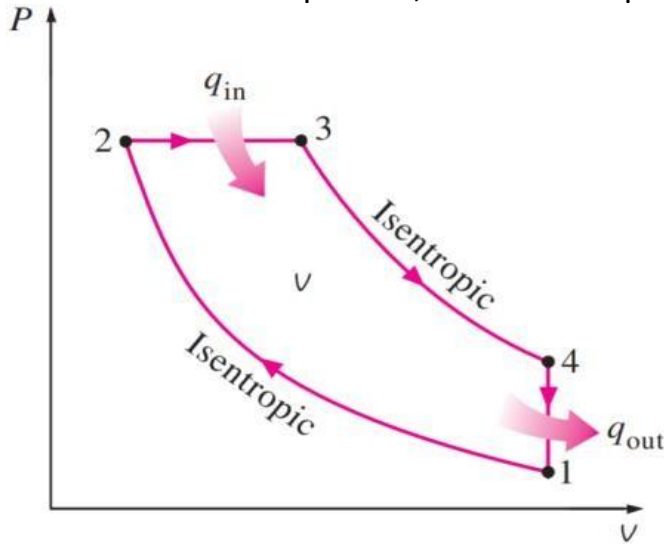
Since process, 1-2 is isentropic,

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^k$$

$$P_2 = 4431 \text{ kPa}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{k-1}$$

$$T_2 = 924.66 \text{ K}$$



(a) P - v diagram

$$Q_{in} = C_p (T_3 - T_2)$$

$$T_3 = T_{max} = 2591.33 \text{ K}$$

$$\frac{P_2 v_2}{T_2} = \frac{P_3 v_3}{T_3}$$

$$\frac{v_2}{v_3} = r_c = 2.8$$

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3} \right)^{k-1}$$

$$T_4 = 1325.37 \text{ K}$$

Heat rejected, $Q_{out} = C_v (T_4 - T_1) = 726.88 \text{ kJ/kg}$

The net work output, $W_{net} = Q_{in} - Q_{out} = 948.12 \text{ kJ/kg}$

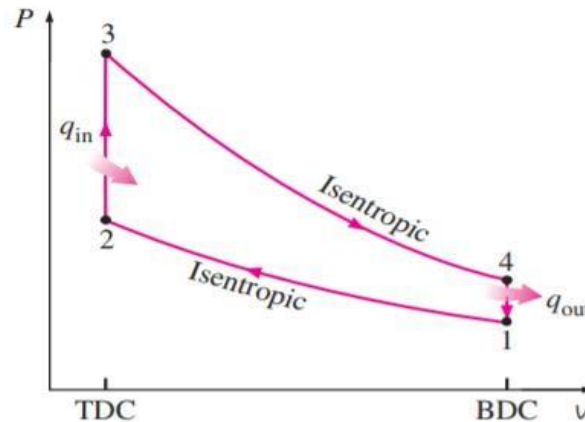
Thermal efficiency, $\eta_{th} = W_{net} / Q_{in} = 0.566 = 56.6 \%$

The mean effective pressure = $W_{net} / (v_1 - v_2) = 1131.4 \text{ kPa}$

Problem 3

The compression ratio in an air-standard Otto cycle is 10. At the beginning of the compression stroke, the pressure is 0.1 MPa and the temperature is 15°C. The heat transfer to the air per cycle is 1800 kJ/kg air. Determine

1. The pressure and temperature at the end of each process of the cycle.
2. The thermal efficiency.
3. The mean effective pressure.



Problem 4

An ideal Diesel cycle with air as the working fluid has a compression ratio of 18 and a cutoff ratio of 2. At the beginning of the compression process, the working fluid is at 14.7 psia, 80°F, and 117 in³. Utilizing the cold-air-standard assumptions, determine (a) the temperature and pressure of air at the end of each process, (b) the net work output and the thermal efficiency, and (c) the mean effective pressure.



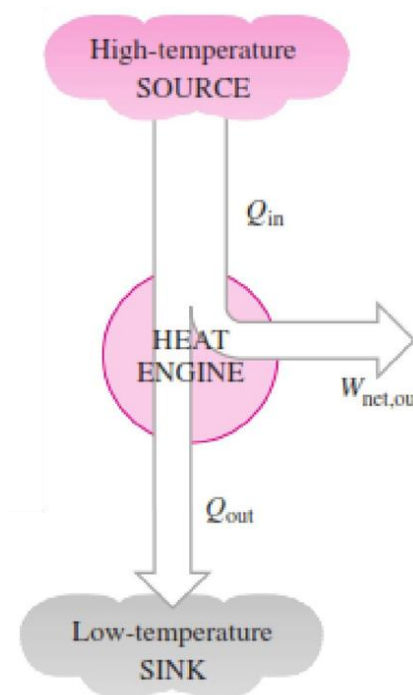
Refrigeration Systems



Heat Engine

- Heat engine is a system that operates in a thermodynamic cycle & does a certain amount of net positive work as a result of heat transfer from a high temperature body to a low temperature body.

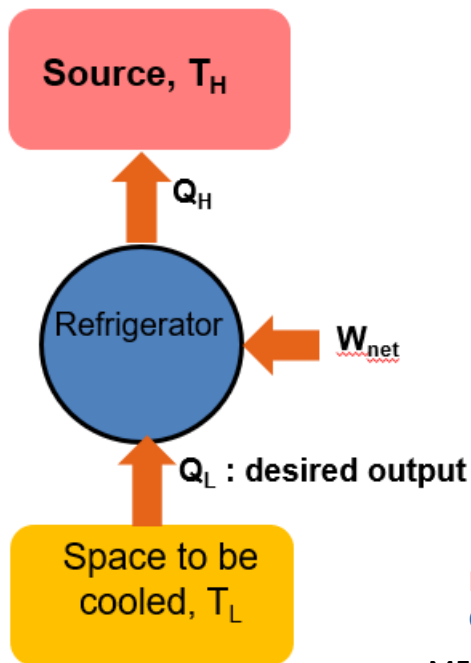
Part of the heat received by a heat engine is converted to work, $W_{net,out}$, while the rest is rejected to a sink.



Refrigeration System

Refrigeration: Refrigeration is the process of removing heat from an enclosed space, or from a substance.

Purpose of Refrigeration: The primary purpose of refrigeration is **lowering** the temperature of the enclosed space or substance and then **maintaining** that lower temperature.



$$COP = \frac{\text{Desired output}}{\text{Required input}}$$

$$COP_R = \frac{Q_L}{W_{net}} = \frac{Q_L}{Q_H - Q_L} = \frac{T_L}{T_H - T_L}$$

Reference Book: Thermodynamics: An Engineering Approach, Yunus A. Cengel, Michael A. Boles. Chapter -11 (Initial Portion)



Application of Refrigeration

- **Domestic Refrigeration**
- Commercial Refrigeration
- Industrial Refrigeration
- Transportation/Mobile Refrigeration



Application of Refrigeration

- Domestic Refrigeration
- **Commercial Refrigeration**
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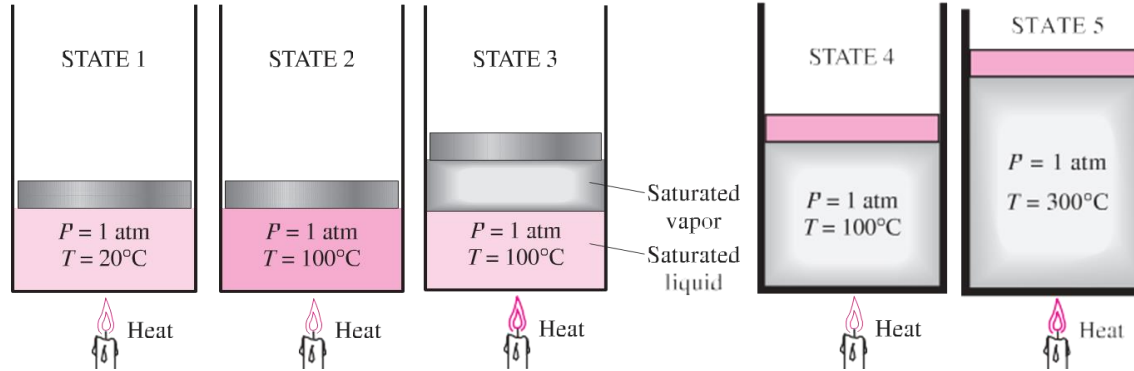


Application of Refrigeration

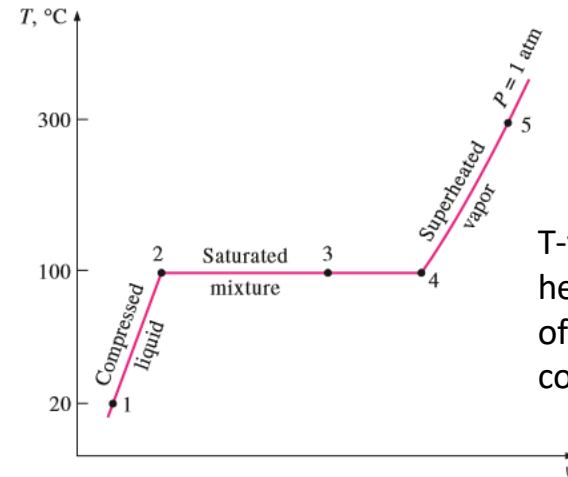
- Domestic Refrigeration
- Commercial Refrigeration
- Industrial Refrigeration
- **Transportation/Mobile Refrigeration**



Vapor-Liquid-Solid Phase Equilibrium



1. Compressed ($P > P_{sat}(T)$) or subcooled liquid ($T < T_{sat}(P)$).
2. Saturated liquid ($T = T_{sat}(P)$).
3. Saturated liquid-vapour mixture ($T = T_{sat}(P)$).
4. Saturated vapour ($T = T_{sat}(P)$).
5. Superheated vapour ($T > T_{sat}(P)$).



T-v diagram for the heating process of water at constant pressure



Vapor-Liquid-Solid Phase Equilibrium

- **Saturation temperature**, T_{sat} is the temperature at which vaporization takes place at a given pressure. This pressure is called the **saturation pressure**, P_{sat} for the given temperature.
- If a substance exists as liquid at T_{sat} and P_{sat} is called a **saturated liquid**.
- If the temperature of the liquid is lower than the saturation temperature for the existing pressure, it is called either a **sub-cooled liquid** (implying $T < T_{sat}(P)$) or a **compressed liquid** (implying $P > P_{sat}(P)$).
- If a substance exists as vapour at T_{sat} , it is called **saturated vapor**. When the vapour is at $T > T_{sat}$, it is said to exist as **superheated vapor**.



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